



Varendra University
Department of Computer Science and Engineering
Course Outline

Course Title: Operating System and System Programming
Course Code: CSE 3203
Course Type: Theory
Credits: 3
Prerequisite Knowledge: CSE 1201

Year and Semester: 3th Year, Summer semester, 2026

Instructor's details:

Name: Md. Fayzul Islam
Designation: Lecturer
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Power Point Slide's and Virtual Learning Platform Web Link: Teams classroom.

Motivation of Course:

Learning Operating Systems and System Programming motivates me to understand how software directly interacts with hardware and manages resources efficiently. It builds a strong foundation for writing high-performance, reliable programs and deepens my problem-solving skills at the system level.

Course Objective:

The objective of this course is to provide a clear understanding of the fundamental concepts of Operating Systems and System Programming. It aims to teach how an operating system manages processes, memory, files, and I/O devices efficiently. The course also focuses on system-level programming concepts such as system calls, process control, and inter-process communication. Students will learn how to design and implement programs that interact directly with the operating system. Finally, the course prepares students to analyze system performance and develop reliable, efficient system software.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

COs	Description	Taxonomy domain/level	POs	WK	WP	EA	Teaching-Learning strategy	Assessment strategy
CO1	Explain the fundamental structure and functions of operating systems	Cognitive/ Understand	1	3			Lectures/Students centered learning	Class Test
CO2	Apply process management and CPU scheduling algorithms to solve scheduling problems	Cognitive/ Apply	3	5			Lectures/Students centered learning	Mid term
CO3	Analyze and implement solutions for concurrency, deadlock, and memory management problem	Cognitive/ Analyze	5	2			Lectures/Students centered learning	Final

Teaching-Learning strategy

COs	Teaching-Learning strategy	Assessment strategy
CO1	Lectures, Power Point Slide, Book	Class Test
CO2	Lecture Slides, Hands on class work	Mid-term exam
CO3	Lectures Slides, Hands on class work	Final Exam

Assessment Detail

Assessment Tools		Marks (%)	
Continuous Assessment (CA)	Class Participation	10%	40%
	Class Test, Presentation, Presentation	30%	
Summative Assessment (SA)	Mid-term Examination	24%	60%
	Final Examination	36%	
Total		100%	

Textbook:

1. A. Silberschatz and Peter Baer Galvin : **Operating System Concepts**, Wiley Publisher
2. Tanenbaum : **Operating Systems**, Prentice Hall

Books Recommended:

1. Madnick and J. Donovan : **Operating Systems**, McGraw-Hill

Course Conducting/Course Policies:

1. It is the student's responsibility to gather information about the assignments and covered topics if he/she does miss the lecture
2. Regular class attendance is mandatory. Points will be taken off for missing classes.
3. Without **50% of** attendance, sitting for final exam is **NOT allowed**.
4. The students must enter the **classroom in time** to get the attendance. **No student** will be allowed to enter the classroom after the attendance has been done.
5. Once the attendance is done, a student can leave the class if he or she thinks that he or she is not getting benefits from the class
6. The reading materials for each class will be available at the **above given link** prior to that class so that student may have a cursory look into the materials.
7. The date and syllabus of quiz/class test will be announced in time in Microsoft Team
8. Students will be **notified** in due time for class cancelation, extra class, make-up class and tutorial class.
9. Students are encouraged to participate in the class discussion and to **ask questions**. The student can ask any question without any hesitation as long as he or she can't understand the topics being discussed; please keep in mind that if you don't understand, it's not your fault, it's my limitation that I could not make you understand. The class is expected to be interactive. The class is expected to be **interactive**.
10. It is expected that the student will also provide some new knowledge related to the curriculum and then make the class as a **place of knowledge sharing among all participants, both teacher and students**.
11. Any attempt for **unfairmeans** in the examination is **strictly prohibited**.

Class & Exam Schedule, Topics and Readings:

Sessions	Topics	Readings
Week-1	Introduction to Operating system, computer System Structure, different types of OS (cont.).	• Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 1) • PowerPoint slides 1
Week-2	User mode-Kernel mode, User Operating System Interface, System Call, Caching	• Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 2) • PowerPoint slides 1

Week-3	Computer System Architecture, Operating System services, System call types, API	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 3) - PowerPoint slides 2
Week-4	Virtual machine, Process, Process control block, context switching, Process creation and termination, inter-process communication.	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 4) - PowerPoint slides 2 & 3
Week-5	Introduction to CPU scheduling algorithm and working principle of different types of CPU scheduling algorithm.	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 6) - PowerPoint slides 4
Week-6	Working principle of different types of CPU scheduling algorithm.	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 6) - PowerPoint slides 4
Week-7	CT1 Topic TBD (To Be Declared)	
Week-8	Process Synchronization, race condition, critical section problem, Solution algorithms to critical section problem algorithms (Synchronization Hardware, Peterson's Solution)	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 7) - PowerPoint slides 5
Week-9	Solution algorithms to critical section problem algorithms (Semaphore, Monitor) and introduction to Deadlock	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 7 & 8) - PowerPoint slides 5 & 6
Week-10	Deadlock with examples, Deadlock necessary conditions, System model, Resource allocation graph, Methods for handling deadlocks, Banker's algorithm	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 8) - PowerPoint slides 6
Week-11	Memory management, logical address vs physical address, Swapping, Contiguous memory allocation (cont.), fragmentation	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 9) - PowerPoint slides 7
Week-12	Paging, Segmentation technique in memory management	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 9) - PowerPoint slides 7
Week-13	Virtual memory, demand paging, page replacement algorithm, cache coherence protocol.	- Silberschatz and Peter Baer Galvin: Operating System Concepts, Wiley Publisher (chapter 10) - PowerPoint slides 8
Week-14	Reviews, Class Test and Final Exam	